ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

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ABSTRACT

Surveillance MRI after treatment for glioblastoma and other aggressive gliomas must be interpreted with prior imaging and the treatment course. A progression score alone does not show the clinician the evidence that produced it or what it leaves out. We built PRISM, a modular system that combines image-derived measurements, treatment context, separate image and timeline pathways, uncertainty and provenance in a locally generated, checked report. On MU-Glioma-Post, the timeline and image pathways reached AUCs of 0.82 and 0.71. Applied unchanged, the combined score reached 0.77 on LUMIERE; at UCSF-ALPTDG, single-scan image measurements reached 0.80 and the image pathway 0.73. The evidence behind the scores differed by cohort: treatment timing at MU-Glioma-Post and enhancing volume at UCSF-ALPTDG. In a second, cross-cohort report evaluation, 406 of 406 reports passed a lexical check against their structured records. PRISM is designed to make evidence inspectable, not to replace clinical judgement; its effect on clinical decisions needs a reader study.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

Glioblastoma and other aggressive gliomas require repeated MRI surveillance after surgery and chemoradiotherapy. At each scan, the clinical team interprets the current images against earlier images and the course of treatment. The decision is hardest soon after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is most common in the first weeks after chemoradiotherapy and more frequent when the promoter of MGMT, a DNA-repair gene, is methylated (Taal et al. 2008; Brandes et al. 2008). The information needed for that judgement is longitudinal: images and their change, treatment timing and the parts of the record that are known on the scan day.

Radiomics and other prediction models can quantify image patterns, but a probability alone does not show the treatment context, the measurements behind an image input, the uncertainty around a score, or what the model did not use. Surveillance data make these gaps hard to close,

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because public datasets often contain images but few treatment dates, centres document progression differently, the hardest situations are uncommon, and segmented volumes need inspection before they are used as inputs. Small cohorts also make a single test split an unstable account of performance (An et al. 2021; Varoquaux 2018). Only 3 of 20 studies in a recent systematic review of automated response assessment included external evaluation (Shi et al. 2025).

Our initial radiomics work on MU-Glioma-Post led us to ask whether a system could turn the evidence available at a post-treatment scan into a report that a clinician can inspect: one that keeps image and treatment-timeline evidence separate, transfers to other centres without refitting, states where the evidence is thin and is written inside the hospital.

PRISM, the system we built, turns each scan into a structured evidence record: clinical and treatment-timeline inputs, measurements from the segmented tumour, separate image and timeline scores, a combined score, uncertainty fields and their provenance. A local language model writes the narrative from that record, and a deterministic checker tests the narrative against it. We developed PRISM on MU-Glioma-Post (MU) and applied its models unchanged to LUMIERE and UCSF-ALPTDG (UCSF). The component models ranked scans at the external centres, but the inputs that drove a score differed by cohort, so the report shows the two pathway scores separately, each with the inputs it is based on.

2. Methods

2.1. PRISM

PRISM handles each scan in five modules (Figure 1) and writes their outputs to one structured evidence record. The timeline pathway and the image pathway keep the two kinds of evidence apart, so that the image pathway can run where no treatment dates exist and the audit can show which evidence a score used. Segmentation masks supply the measurements and images displayed in the report. The record also carries a set of plausible outcomes, agreement between pathways, how many scans like this one have been tested and the provenance of every displayed value. The report is written locally from that record, so patient data do not leave the hospital.

The timeline pathway scores a scan from clinical data and the patient's position in treatment. On MU its inputs are age, sex, IDH mutation and MGMT promoter methylation, the days since diagnosis and since surgery, the start, end and dose of radiotherapy, chemotherapy and the visit index, each taken as of the scan day. The first four alone form the clinical-only reference model. Where a cohort records no treatment dates, PRISM uses a reduced timeline that keeps the clinical inputs, the days since surgery and a flag for the first weeks after radiotherapy. The flag comes from the rater's note at LUMIERE and is approximated from the start of radiotherapy at UCSF. Both pathways are ridge logistic regressions; scaling and any selection were fitted on the training data only. An unknown IDH or MGMT status is coded as the reference category, and a missing measurement is filled with the training median, with a missing-value indicator, inside the fitted pipeline.

The image pathway scores a scan from simple measurements of the segmented tumour. On MU its 27 inputs are the volumes, shape and relative intensity of the segmented compartments at the scan, their change from the previous scan, and the enhancing and total lesion volumes relative to the smallest in the patient's history ([Supplementary section: image pathway inputs and their harmonisation at LUMIERE and UCSF]). It holds no dates, intervals or treatment flags, because

such inputs carry when a scan was taken, which the timeline already holds, and the image pathway has to run where no dates exist (Section 3.3). A single-scan version keeps only the 19 measurements of the current scan.

The combined score is one ridge model fitted on the inputs of both pathways and is shown beside them. At LUMIERE and UCSF its inputs are the reduced timeline, the harmonised volumes and their change, the interval since the previous scan and whether there was one.

The measurements come from the segmentation masks supplied with each cohort, which are the automatic masks released with MU (Mahmoud et al. 2025), the DeepBraTumIA masks of LUMIERE (Suter et al. 2022) and the masks provided with UCSF (Fields et al. 2024). PRISM renders the report's figure from the same masks by code.

Each scan's confidence has three parts, a conformal set, a pathway-agreement statement and a reliability card (Mitchell et al. 2019). The conformal set is a patient-level split conformal prediction set at the 80% level (Angelopoulos and Bates 2023), built to contain the true outcome for at least 80% of scans; it holds one outcome when the score is clear and both when it is not. It comes from the combined model refitted on 70% of the MU patients and calibrated on the remaining 30% ([Supplementary section: system results]). On MU this split is made inside each training fold. For the external cohorts we applied one MU-calibrated set unchanged, which assumes that their scans are exchangeable with MU's. The two pathways agree when both scores lie above, or both below, each pathway's median on MU, so the rule needs no labels and can be applied to a single scan. The medians come from the MU training scans of the locked models, or, on MU, from the held-out scores of the other folds. The reliability card describes the scan's situation, its phase of care crossed with the change in enhancing tumour since the previous scan, by looking it up in a table of that cohort's held-out results. The card is a record of measured results, so the scan's own result is part of its row. A situation with at least 10 progression and 10 non-progression scans is tested, and the card gives its AUCs. A situation with fewer has few cases, and the card gives counts only; a situation with none is not tested.

A local language model writes the report from the scan's record, and a deterministic checker tests the report against that record. The model (Qwen3.8-27B) runs offline on one GPU at temperature 0 and receives only the record; the checker rejects any number, flag or confidence statement that is not in the record, any omission of required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

In all three cohorts the primary population is grade 4 glioma. We fitted each model once on MU and applied it unchanged to LUMIERE and UCSF, and we call these models locked. Table 1 lists what each cohort records; UCSF holds one pair of consecutive post-treatment scans per patient, and because the first scan has no earlier scan we evaluated the second only. Progression is documented in each cohort's own record, as a dated clinical decision at MU, as an expert rating of response at each scan at LUMIERE and as a progression date supplied with the dataset at UCSF.

The primary task is to decide, at each scan, whether this is the scan at which progression is documented. On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. At

UCSF a positive is a second scan whose progression date lies within 30 days of it and closer to it than to the first scan; a negative is a second scan followed by at least 120 days without progression. On MU the intended use is scans from the start of radiotherapy onwards, and unless stated MU results are for this population; LUMIERE and UCSF are evaluated on all scans in their detection sets. We divide the intended use into three phases of care: during radiotherapy, early after radiotherapy (up to 12 weeks after its end) and later. A secondary task, predicting progression within 120 days of a scan, is reported in the supplement.

2.3. Evaluation

On MU, every model was evaluated by repeated patient-grouped cross-validation. We used five repeats of five folds, with all scans of a patient in the same fold, and report the mean over repeats of the pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws. For the models and the conformal sets, no patient appears in more than one of fitting, calibration and testing. We chose repeated splits because an earlier radiomics model on MU reached an AUC of 0.80 on one test split and averaged 0.72 when re-selected and re-tested over 500 splits of the same design (Supplementary Section S8 and Figure S2).

For the external tests, the locked models, fitted on all grade 4 detection scans of MU from every phase of care, were applied under plans committed before any model score was compared with an external label. We made one deviation: the UCSF-ALPTDG plan named all MU detection scans for fitting, but we fitted on the grade 4 scans. Fitting on all gliomas gave almost the same AUC at LUMIERE (0.77); at UCSF we tested the all-glioma models on all-glioma scans ([Supplementary section: deviations from the plans]). External AUCs have 95% CIs from 1,000 patient-level bootstrap draws. As a reference, we also fitted each UCSF model on UCSF itself by repeated cross-validation. To test whether deep networks see more in the images than the measurements do, we fine-tuned segmentation networks to score each scan directly, against a control trained on permuted labels ([Supplementary section: deep networks at MU and UCSF]). Table 2 lists the planned questions and their answers.

Results of analyses not fixed in advance carry a dagger (†) where they first appear in a paragraph, legend or table, and the analyses are listed in [Supplementary section: analyses not fixed in advance]. Among them are the definition of the image pathway on MU, whose use at UCSF and in the system tests was planned, and the agreement rule and the reliability card, which were defined after the cohort results had been seen. We use comparisons within a phase of care or within an enhancement category to diagnose what a score depends on; a within-phase AUC pools only the pairs of a progression and a non-progression scan from the same phase.

3. Results

3.1. The pathways provide distinct evidence at the development centre

In the intended use on MU (146 scans; Table 1), the image pathway† detected the progression scan with an AUC of 0.71 (95% CI 0.63–0.78), and the timeline pathway with 0.82 (0.75–0.90), mostly by separating phases of care, within which it did no better than the images (Section 3.3). Clinical data alone, without the timeline, scored 0.53, close to chance. The combined score that PRISM shows, fitted on the inputs of both pathways†, also reached 0.82, so on MU the images did not raise the

timeline's AUC. A planned model that adds image measurements and three schedule inputs to the timeline fell short of its target gain of 0.03 (Table 2; Section 3.3). Predicting progression within 120 days of a scan was harder; the model reached 0.60 (0.52–0.69) in the intended use (Supplementary Section S23). A timeline pathway needs treatment dates, which the two external cohorts largely lack, so we next tested how each pathway transfers to them.

3.2. The components rank scans at two external centres without refitting

The locked combined model reached 0.77 (0.68–0.86) on LUMIERE (Figure 2) and met its planned transfer rule (Table 2). At LUMIERE the combined score was above either pathway alone, images 0.65 and the reduced timeline 0.69† (the gain over the timeline alone had an interval that included zero). Clinical data alone stayed close to chance in all three cohorts (at LUMIERE†; Figure 2). Enhancing tumour volume alone, with no fitted model†, scored 0.62 on MU, below the image pathway, and 0.66 on LUMIERE, level with the image pathway and the timeline (Figure 2).

At UCSF, both planned image models transferred. Measurements of the single scan reached 0.80 (0.69–0.89), and the image pathway, which adds change from the previous scan, reached 0.73 (0.60–0.84), meeting its planned transfer rule (Table 2). With coefficients fitted on MU, the change inputs therefore transferred less well than the measurements of the scan itself (paired difference −0.07, −0.13 to −0.01). Training at UCSF raised the image pathway to 0.86 (difference 0.14, 0.03–0.26), while the single-scan model scored about the same when trained there†. The reduced timeline had the lower AUC of the two pathways, 0.56; UCSF can only approximate its flag for the early weeks after radiotherapy (Section 2.1). The locked combined score reached 0.86 (0.76–0.93). It uses the approximated flag, and part of its AUC comes from the interval between the two scans, which was shorter before progression† (Figure 2).

Read as a risk, the combined score was calibrated in the large at LUMIERE† but overpredicted progression at UCSF (observed over expected 0.72, 0.55–0.91), and a threshold set at the MU median† called 90% of UCSF scans positive. A new centre therefore needs to recalibrate the score before reading it as a probability or applying a threshold ([Supplementary section: calibration and thresholds]).

3.3. The evidence behind a score differs by cohort

At MU the scores depended mainly on where the patient was in treatment, and at UCSF on how much the tumour enhanced† (Figure 3). At MU the scan schedule alone separates most progression scans from most non-progression scans, because most progression scans came more than 12 weeks after radiotherapy and most non-progression scans came within those 12 weeks (Table 1). Across phases of care, adding the timeline to clinical data and images gained 0.11 (0.04–0.20); within a phase it gained −0.02 (−0.13 to +0.11) (Figure 3A). Within a phase of care, where the schedule cannot separate the scans, the image pathway gave about as much evidence as the timeline, with AUCs of 0.68 (0.58–0.77) and 0.64 (0.51–0.78).

Adding inputs to the image model one step at a time showed which of them raised its AUC on MU† (Figure 3C). Adding change from earlier scans to the current scan's measurements gave the image pathway, 0.05 higher with an interval that included zero (−0.003 to +0.106). The third step added the three schedule inputs of the planned model of Section 3.1: the interval since the previous scan, a flag for whether there was one, and the smallest volume counted from the end of

radiotherapy. They raised the AUC by a further 0.10 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04).

At UCSF the progression calls followed enhancement. Enhancing tumour volume alone, the simplest baseline, gave an AUC of 0.88 (0.79–0.95)†, higher than the point estimates of the locked image models; UCSF rated enhancement as increased at 27 of the 29 progression scans, and progression was dated a median of 7 days after the evaluated scan (Figure 3B). Among the 37 scans rated as showing increased enhancement, the single-scan image model reached 0.65 (0.44–0.84), so beyond that rating the measurements added little that this sample could detect. At UCSF a model therefore scores well largely by agreeing with a progression call that was itself made from enhancement at the same scan, which is what the image pathway measures. Deep networks fine-tuned on UCSF scans learned above a permuted-label control (Table 2), and their scores tracked enhancing volume† ([Supplementary section: deep networks at MU and UCSF]).

3.4. The confidence keeps its coverage on average, and disagreement marks the less accurate scans at MU

The 80% conformal sets of the combined score covered 78% to 87% of scans in the three cohorts, at or above the planned threshold of 78% (Figure 4A). Most sets were informative, holding a single outcome for 62% to 75% of scans. At UCSF the point estimate sat at the threshold and its interval reached below it (68% to 87%). Coverage was uneven by outcome†: 77% (67% to 87%) of progression scans at LUMIERE against 94% of non-progression scans, and 65% (51% to 79%) of non-progression scans at UCSF against 97% of progression scans; on MU the two were 82% and 83%. Thresholds fitted for each outcome on MU did not correct this ([Supplementary section: system results]).

Disagreement between the pathways marked the less accurate scans at MU but not, under the planned rule, at the external centres. On MU the pathways disagreed on 40% of scans, and the combined score's AUC was 0.90 where they agreed and 0.59 where they disagreed; setting the disagreeing scans aside also gained 0.08 (0.05–0.12) more than removing the same number of scans matched for phase of care. Both results met their planned tests (Table 2; Figure 4B). At LUMIERE and UCSF the planned tests, which took each cohort's medians from its own scores, were not met (at LUMIERE, 0.11, −0.05 to 0.27).

With the MU medians that the system uses, because they can be applied to a single scan, disagreement did mark the less accurate scans at LUMIERE†. The pathways disagreed on 60% of LUMIERE scans and 48% of UCSF scans. At LUMIERE the AUC was 0.89 where they agreed and 0.67 where they disagreed (0.22, 0.10–0.34), and setting the disagreeing scans aside gained 0.12 (0.05–0.19) over removing the same number at random; at UCSF the intervals included zero. A conformal set holding both outcomes, used as a flag, met one of the two planned tests in each cohort (Table 2; [Supplementary section: system results]).

Only four situations on the reliability card† reached the tested tier, one at MU and three at LUMIERE. Because LUMIERE records no phase of care, its situations differ by enhancement only. Of the 146 MU scans, 54 fell in the tested situation, most LUMIERE scans did, and every situation at UCSF had few cases (Figure 4C). At MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. Tested situations had lower error than few-case situations at LUMIERE, with overlapping intervals (Table 2; intervals in the supplement); at MU the order was reversed, because the few-case scans there were mostly non-progression scans early after radiother-

apy, which the models score easily.

3.5. The report passes an automatic check against its structured record

Every report written by the local model passed the check of its numbers, flags and wording against its structured record, 114 reports at MU, 162 at LUMIERE and 130 at UCSF, 406 in total (Figure 5B). All 220 reports whose situation had few cases or was untested said so, and when a field of the record was changed, 28 of 30 reports followed the change, against a planned minimum of 27 (Table 2). These results follow one revision of the prompt†. Before it, every LUMIERE report failed, because a fixed sentence about the rater's three-month flag quoted a number held in no field. We corrected the prompt, left the checker unchanged and evaluated again on patients not used for the revision ([Supplementary section: report revision]). The check is lexical and cannot judge emphasis, order or what a clinician would infer.

4. Discussion

PRISM is a system for making longitudinal surveillance evidence inspectable. It gives the clinician separate image and treatment-timeline evidence at each scan, the measurements and context behind those fields, and a report written inside the hospital that states where the evidence is thin. The component models ranked scans at two external centres without refitting, but their evidence differed by setting. At the development centre the timeline provided most of the evidence, largely because of when patients were scanned; within a phase of care the images provided about as much. At UCSF enhancing volume provided the score, and enhancing volume alone† did as well as the image pathway trained there. The ranking transferred, and the risk scale did not transfer to UCSF, so a group deploying PRISM at a new centre would recalibrate a probability and set any threshold on local scans.

Because its modules are separate, PRISM can run where treatment dates are missing. The image pathway needs only the scans and their masks, which is what most public surveillance datasets hold; at LUMIERE and UCSF it ran without full treatment dates. Where dates exist, the timeline is added as a second pathway, and the combined score is shown beside both without changing the image model. The reliability card is a lookup table, so a new cohort adds rows to it, and a situation's tier can change as evidence accumulates without refitting anything.

Consider, as an illustration, a scan eight weeks after radiotherapy in which the enhancing tumour has grown. At MU, 1 of the 9 scans in this situation on the card (all grade 4 detection scans) was progression†. The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. PRISM's report would show both scores, state that they disagree, give a conformal set that holds both outcomes and record that the card has few cases for this situation. The clinician then sees that the system has too few tested cases to support a call at this scan, in the weeks when pseudoprogression is most common, and decides with information PRISM does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each of PRISM's components has been described before, but we are not aware of a system that combines them at a single scan, with each kind of evidence kept apart and the system's own test results attached. Automatic segmentation of post-treatment glioma with measurement of change

between scans is established (Rudie et al. 2022), conformal prediction has been applied to glioma segmentation (Elyassirad et al. 2025), clinical and imaging data have been fused to separate pseudoprogression from progression (Gomaa et al. 2025), and model cards report performance by subgroup (Mitchell et al. 2019). The closest system we found combines treatment variables extracted by a language model with tumour volumes to assign a structured follow-up category at one centre, without an uncertainty attached to the output (Jabal et al. 2026). On MU Christodoulou et al. (2026) predicted a label assigned at the patient level, on LUMIERE [cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified] classified each scan into four response categories, and Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres; none of these is detection at the scan.

PRISM was developed at one centre (Table 1), and the two external tests are its only evidence that its ranking of scans transfers. The label is progression as each centre documented it; no scan was re-read for this study, and at UCSF most progression scans were also rated as showing increased enhancement. Subgroups are small, and the analyses marked † need testing on new data. No clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected progression at the scan (AUC 0.60 in the intended use).

The next step is to pool the rare situations, early progression after radiotherapy and growing enhancement without it, across cohorts and train on them. The Burdenko and UCSD-PTGBM datasets add radiotherapy detail and advanced sequences, and models fine-tuned on these long-tail cases could be tested on their scans in the card's thin situations. A reader study in which clinicians use PRISM's reports at the scan would show whether separate evidence and a stated confidence change their decisions.

Conclusion

PRISM keeps image and treatment-timeline evidence apart and shows the clinician both scores at every scan with how far each has been tested there; its models ranked scans at two other centres without refitting.

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Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight mod-

els in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 2. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

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Tables

Table 1. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 scans with a 120-day label (median 2)	Up to 10 scans with a 120-day label (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	Days from surgery only; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no scan carrying a 120-day label are excluded. Reports were evaluated on the 114 intended-use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 3.5).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 2. Questions fixed in advance and their answers. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Result	Met
Detection on MU-Glioma-Post			
Do image measurements add to the timeline?	Planned model $\geq$ 0.75 (lower bound $\geq$ 0.65) and gain $\geq$ 0.03 with CI above 0	Model 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)	No
Do fine-tuned deep networks add?	Gain $\geq$ 0.03 with CI above 0; permuted-label control covers 0.5	No network added; control 0.483 (0.436–0.527)	No (control valid)
Does the locked model transfer to LUMIERE?	External AUC $\geq$ 0.65	AUC above 0.65 (Section 3.2)	Yes
UCSF-ALPTDG			
Does the image pathway transfer?	AUC $\geq$ 0.65, lower bound $>$ 0.5	Both conditions met (Section 3.2)	Yes
Do images add to clinical data without a timeline?	Gain with CI above 0	+0.248 (+0.096 to +0.392)	Yes
Does training at UCSF-ALPTDG do better than the locked model?	Descriptive	Image pathway higher when trained there; single-scan model about the same (Section 3.2)	Descriptive
Do deep networks learn at UCSF-ALPTDG?	Above the permuted-label control; control covers 0.5	+0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)	Yes
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)			
Do conformal sets keep their coverage?	Coverage $\geq$ 0.78 at the 80% level in each cohort	Combined score met on point estimates in all three, but not for every outcome (Section 3.4); images at LUMIERE 0.708 and timeline at UCSF-ALPTDG 0.696	Yes (combined)
Do flags pick out less accurate scans?	AUC unflagged minus flagged $>$ 0, CI above 0	Disagreement at MU-Glioma-Post +0.307 (+0.175 to +0.432); per-cohort results in the supplement[a]	MU: disagreement and set; LUMIERE: set; UCSF: neither
Does setting flagged scans aside beat matched removal?	Retained AUC minus matched removal $>$ 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); per-cohort results in the supplement[a]	MU: disagreement; LUMIERE: neither; UCSF: set
Do tested situations have lower error?	Descriptive: tested below few cases in Brier score and error	Brier 0.187 vs 0.258 at LUMIERE; reversed at MU-Glioma-Post (Section 3.4)	Descriptive
Are reports faithful to the record?	$\geq$ 95% pass the checker in each cohort	First evaluation failed at LUMIERE; second evaluation: all passed (Section 3.5)	No, then yes
Is the tier stated where evidence is thin?	100%	All (Section 3.5)	Yes
Do reports follow a changed record?	$\geq$ 27 of 30 perturbed records	28 of 30 (Section 3.5)	Yes
Smallest report model passing the checker rule in every cohort	Descriptive	Qwen3.8-27B (supplement)	Descriptive

Plans were committed before each analysis: detection (commit 2b4c927), UCSF-ALPTDG (0e7c9fe), the system (e7cba80) and the second report evaluation (4f7e7b9). In the first row, the planned model gained +0.041 (+0.006 to +0.081) over all scans of all gliomas, which may reflect its schedule inputs (Section 3.3). Matched removal keeps the phase-of-care mix of the removed scans (random removal at LUMIERE, which has no phase); "set" is a conformal set holding both outcomes. The flag and tier tests were fixed in advance, but the agreement rule and the card tiers were defined after the cohort results had been seen[†].

[a] The Met cells use each cohort's own medians, as fixed in advance. With the medians fixed on MU, which the system uses (Section 2.1), disagreement also met both tests at LUMIERE[†]. [Supplementary section: system results]: both tests for each flag in each cohort, with both median rules and with the rule that flags any scan.

Figure legends

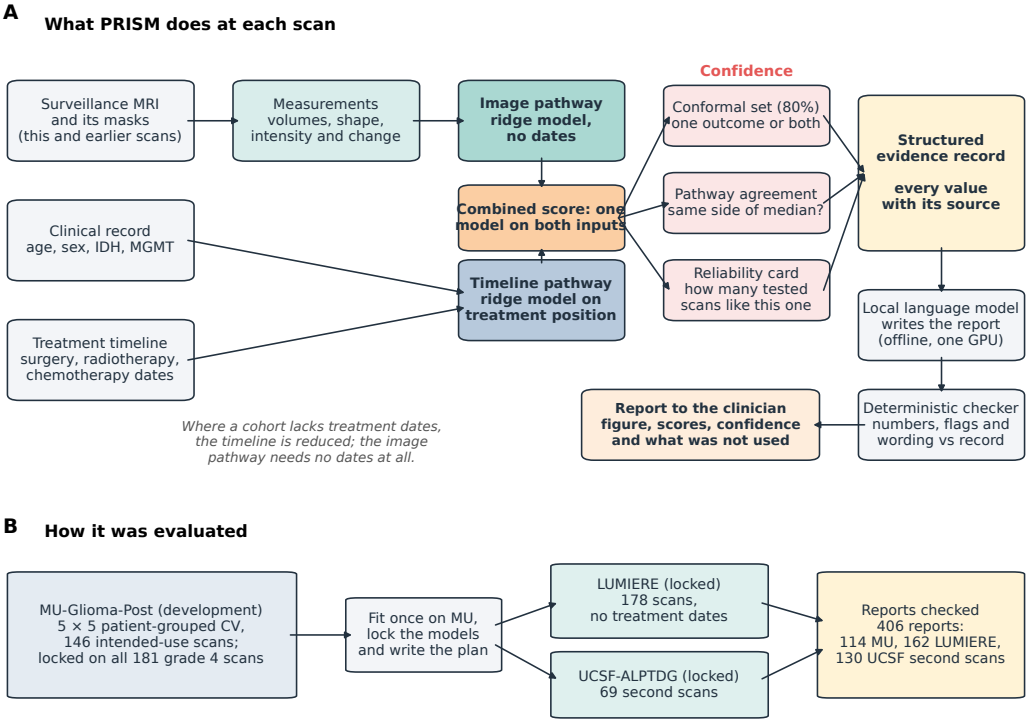


Figure 1. PRISM and how it was evaluated. (A) At each scan, measurements from the segmentation masks feed the image pathway, and clinical data with the patient's position in treatment feed the timeline pathway; the combined score, one ridge model on both sets of inputs, is shown beside them. The confidence has three parts: an 80% conformal prediction set, a statement of whether the two pathways agree, and the reliability card's tier for the scan's situation. Every value enters a structured record with its source; a local language model writes the report from that record, and a deterministic checker tests its numbers, flags and wording against it. (B) Models were developed on MU-Glioma-Post by repeated cross-validation, fitted once and applied unchanged to LUMIERE and UCSF-ALPTDG under plans committed in advance; reports were checked on records from all three cohorts.

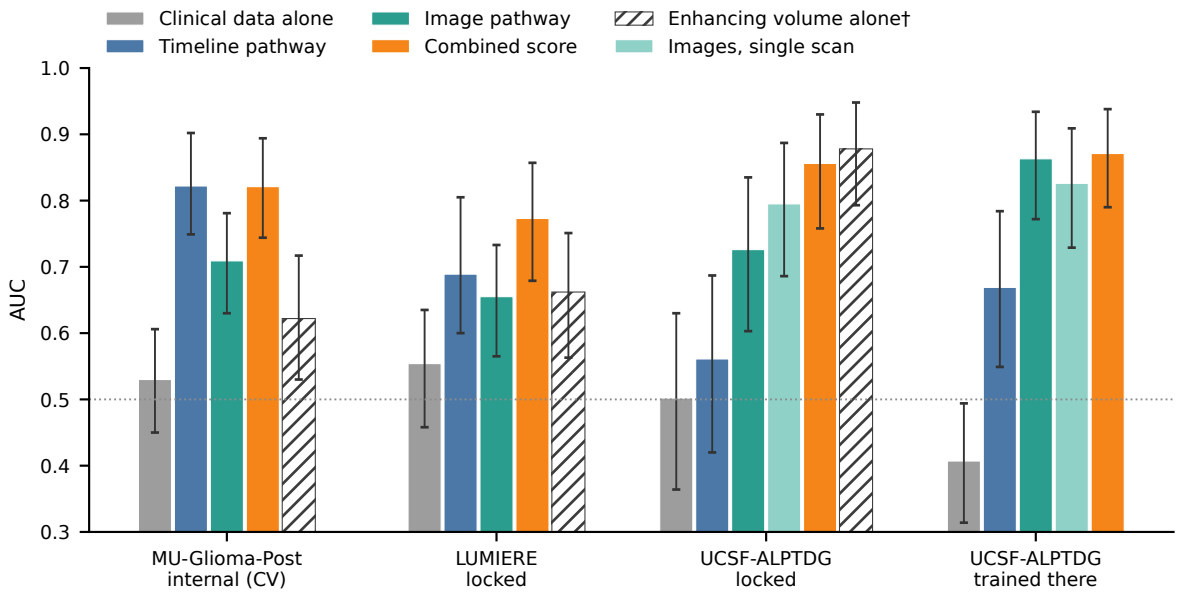


Figure 2. The models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented, by pathway (clinical data alone, timeline, images, combined) and cohort: MU-Glioma-Post (repeated cross-validation, intended use); LUMIERE and UCSF-ALPTDG (models fitted once on MU-Glioma-Post and applied unchanged); and UCSF-ALPTDG with models trained there, as a reference, where clinical data alone fall below 0.5, as a model with no signal can under cross-validation. At LUMIERE and UCSF-ALPTDG the timeline is the reduced version; the clinical-only model at LUMIERE was not fixed in advance. At UCSF-ALPTDG the images are shown for the single-scan model and for the image pathway with change features, which is the system's image pathway. The hatched bar is the log enhancing tumour volume of the scan alone, with no fitted model[†]. Error bars are 95% patient-bootstrap CIs; the dotted line is chance.

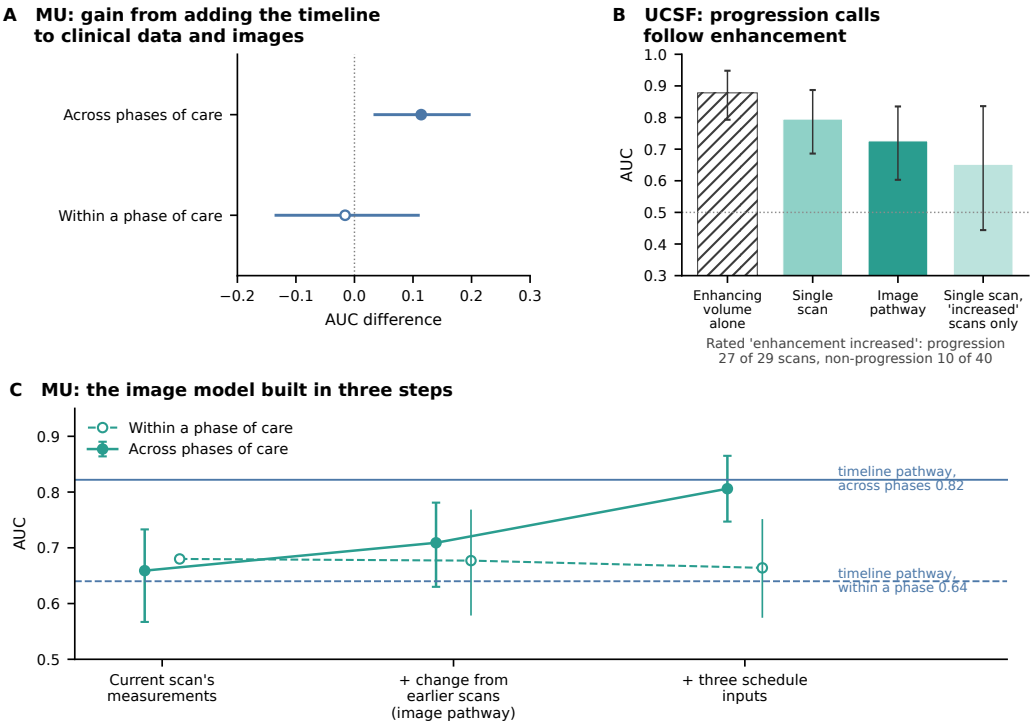


Figure 3. What each score rests on depends on the cohort. (A) MU-Glioma-Post: the gain from adding the timeline to clinical data and images, across phases of care and within them. (B) UCSF-ALPTDG: AUC of enhancing tumour volume alone beside the locked image models, with the counts of progression and non-progression scans that UCSF rated as showing increased enhancement noted below the panel. (C) The image model on MU-Glioma-Post built in three steps (current scan; with change from earlier scans; with schedule inputs), across and within phases of care; the interval of the second step includes zero. The audits in all panels were not fixed in advance (†); the locked UCSF-ALPTDG model AUCs in (B) were planned, and the last bar of (B) is restricted to the scans rated as showing increased enhancement. Bars are 95% patient-bootstrap CIs.

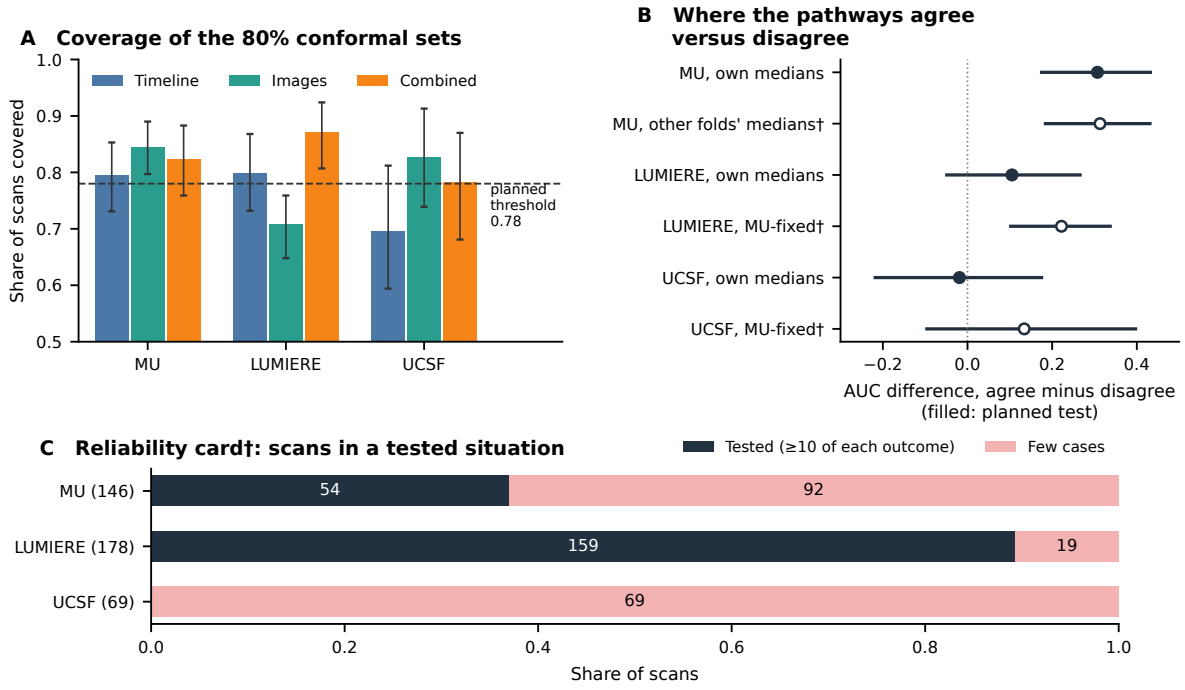


Figure 4. The confidence keeps its coverage on average, and disagreement marks the less accurate scans at MU-Glioma-Post. (A) Coverage of the 80% conformal sets by cohort and pathway; the dashed line is the threshold fixed in advance, met by the combined score on point estimates in all three cohorts; by outcome, coverage fell short for one outcome at LUMIERE and UCSF-ALPTDG (Section 3.4). (B) The combined score's AUC where the pathways agree minus where they disagree, by cohort, with each cohort's own medians (filled, as planned) and with the medians the system uses, fixed on MU-Glioma-Post (open†; on MU-Glioma-Post, the other folds' medians). (C) Scans whose situation (phase of care by change in enhancing tumour) reached the card's tested tier, by cohort; the rest have few cases, and none of these scans was in an untested situation.

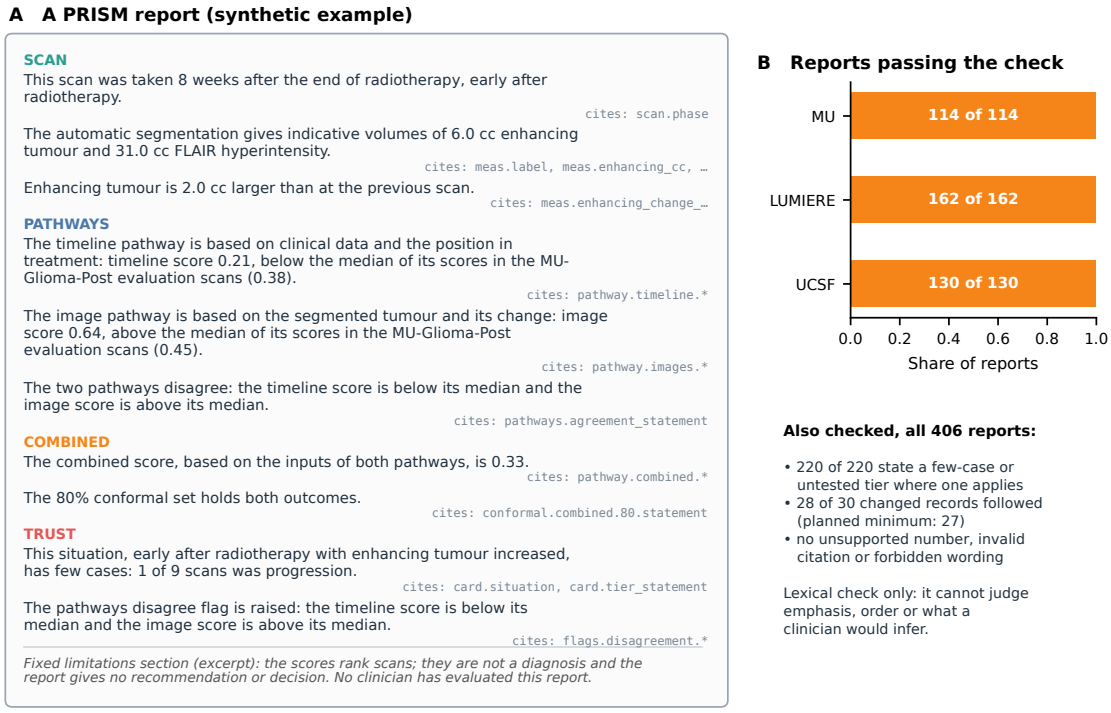


Figure 5. The report states the evidence and passes a check against the record. (A) The narrative of a PRISM report for a synthetic scan taken eight weeks after radiotherapy with growing enhancement, the situation of Section 4; the volumes and scores are invented, and the card row is MU-Glioma-Post's (all grade 4 detection scans). Each sentence lists the record fields it cites, which the checker uses; the delivered report also holds the segmentation figure, tables of every value and a fixed limitations section. (B) Reports passing the lexical check of numbers, flags and wording, after one revision of the prompt following a failure at LUMIERE (Section 3.5).